Competing Risks and Multi-State Models

Concepts, methods and software

Course given at *Karolinska Institute*

May 5-8, 2026

Ronald Geskus[1] and Hein Putter[2]

[1] Oxford University Clinical Research Unit (OUCRU)

Hospital for Tropical Diseases, Ho Chi Minh City, Viet Nam

[2] Department of Biomedical Data Sciences

Leiden University Medical Center, Leiden, the Netherlands

Part III

Estimation: Subdistribution Approach

Contents:

Outline

Subdistribution

- Subdistribution hazard
- Product-limit estimator
- Confidence intervals based on PL-estimator

Log-rank tests

Proportional subdistribution hazards

- Standard errors

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Subdistribution

- Subdistribution hazard
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Notation and definitions

- Competing risks: type $E \in \mathcal{K}$.
  We can assume $\mathcal{K} = \{1, \dots, K\}$
- $T \sim F$ time to event (of any type); $F(t) = \mathrm{P}(T \leq t)$
- Overall hazard h:
  $\overline{F}(t) = 1 - F(t) = \mathrm{P}(T > t) = \exp\{-\int_0^t h(s)ds\}$
- Cause-specific cumulative incidence:
  $F_k(t) = \mathrm{P}(T \leq t, E = k)$; $E \in \mathcal{K}$
  Property: $\sum_{e=1}^{K} F_e(t) = \sum_{e=1}^{K} \mathrm{P}(T \leq t, E = e) = F(t)$
- Subdistribution random variable $T_k \sim F_k$:
  $T_k = T \times I\{E = k\} + \infty \times I\{E \neq k\}$ (event of interest)
- $\mathrm{P}(T_k \leq t) = \mathrm{P}(T \leq t, E = k) = F_k(t)$
- Subdistribution hazard (continuous distribution):

$$h_k = \lim_{\Delta t \downarrow 0} \frac{1}{\Delta t} \times \mathrm{P}\{t \leq T_k < t + \Delta t \mid T_k \geq t\}$$

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$$h_k = \lim_{\Delta t \downarrow 0} \frac{1}{\Delta t} \times \mathrm{P}\{t \leq T_k < t + \Delta t \mid T_k \geq t\}$$

$$\overline{F_k}(t) = \mathrm{P}(T_k > t) = \exp\left\{-\int_0^t h_k(s)ds\right\}$$

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Subdistribution hazard

$$h_k(s) = \lim_{\Delta s \downarrow 0} \frac{1}{\Delta s} \times \mathrm{P}\{s \leq T_k < s + \Delta s \,|\, T_k \geq s\}$$
$$= \lim_{\Delta s \downarrow 0} \frac{\frac{1}{\Delta s}\mathrm{P}\{s \leq T < s + \Delta s, E = k\}}{\mathrm{P}\{T \geq s \text{ or } (T < s, E \neq k)\}}$$

- Denominator: event free or with earlier competing event
- Interpretation controversial
  - Not a rate in epidemiological sense, unless we can include immune individuals in the risk set (e.g. vaccination; unobserved cure)
- One-to-one relation with crude risk

$$\overline{F_k}(t) = \prod_{t_l \leq t} \left\{1 - h_k(t_l)\right\} \quad \text{or} \quad \overline{F_k}(t) = \exp\left\{-\int_0^t h_k(s)ds\right\}$$

Estimation

week	0-1	1-2	2-3	3-4	4-5	5-6	6-7	>7
infection	1	2	6	11	9	11	2	18
cumulative	1	3	9	20	29	40	42	60
discharge	5	9	6	6	9	12	4	35
cumulative	5	14	20	26	35	47	51	86

- Subdistribution. Estimated as frequency of events:

  $\widehat{P}$(infection $\leq$6 weeks)$=40/146$

  $\widehat{P}$(discharge $\leq$6 weeks)$=47/146$

  Individuals with competing event remain in denominator, competing event ignored in estimation

Observed data

$\{(v_1, x_1, e_1\delta_1), \ldots, (v_N, x_N, e_N\delta_N)\}$
Note: $t_i < c_j < v_{j'}$ in case of ties

- $x_i = \min\{t_i, c_i\}$, $\delta_i = \{t_i \leq c_i\}$, $e_i \in \{1, \ldots, K\}$
- $t_{(i)}$ ordered unique event times of any type
- $d_k(t_{(i)})$ number of events at $t_{(i)}$ of type k
- $d(t_{(i)})$ total number of events at $t_{(i)}$
- $r(t_{(i)})$ number observed at risk
- $r^*(t_{(i)})$ number "in risk set" (for subdistribution hazard)

Estimation of subdistribution hazard

$$\widehat{h_k}(t_{(i)}) = \frac{d_k(t_{(i)})}{r^*(t_{(i)})}$$

Complete data (no censoring/truncation): individuals with competing event remain in risk set forever. Small change in data setup, standard software

Right censored event time: three reasons

Administrative censoring At date of analysis. Still in follow-up and event may be observed in the future

Loss to follow-up Left the study before date of analysis. May happen for different reasons (move to another country, too ill to participate). Event may have happened, but is not observed.

Competing risks Two types:

- Absorbing event: other event (e.g. death) prevents occurrence of event of interest. No further follow-up information
- Non-absorbing event: “artificial censoring”. Still in study, but conditions have changed too much to stay included.

  Another example: first event analysis

Estimation of subdistribution hazard

$$\widehat{h_k}(t_{(i)}) = \frac{d_k(t_{(i)})}{r^*(t_{(i)})}$$

Complete data (no censoring/truncation): individuals with competing event remain in risk set forever. Small change in data setup, standard software

Administrative censoring only: individuals with a competing event leave risk set at their date of censoring. Standard software can be used with small change in follow-up

Estimation of subdistribution hazard

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Complete data (no censoring/truncation): individuals with competing event remain in risk set forever. Small change in data setup, standard software

Administrative censoring only: individuals with a competing event leave risk set at their date of censoring. Standard software can be used with small change in follow-up

Censoring time unknown in some individuals: Estimate time-to-censoring distribution.

Probabilistic setup data

- Event time T_i; may be right censored (administratively, loss to follow-up)
  Censoring time C_i
  $X_i = \min\{T_i, C_i\}$, $\Delta_i = I(T_i \leq C_i)$
- Left truncation (late entry).
  Entry time V_i
- Sample: $\{V_i, X_i, E_i\Delta_i\}, i = 1, \ldots, N$
- Assume (V_i, C_i) independent of/non-informative for (T_i, E_i) (possibly conditional on covariables in regression model)

Estimation of $\overline{\Gamma}$

Simplest: via standard PL-form

- $\widehat{\overline{\Gamma}}$: reverse role of event time T_i and censoring C_i:

$$\widehat{\overline{\Gamma}}(t) = \widehat{\mathrm{P}}(C > t) = \prod_{i:c_{(i)} \le t} \left\{ 1 - \frac{m_i}{r(c_{(i)})} \right\}$$

m_i: number of censorings at $c_{(i)}$

Estimation of subdistribution hazard

$$\widehat{h_k}(t_{(i)}) = \frac{d_k(t_{(i)})}{r^*(t_{(i)})}$$

Complete data (no censoring/truncation): individuals with competing event remain in risk set forever. Small change in data setup, standard software

Administrative censoring only: individuals with a competing event leave risk set at their date of censoring. Standard software can be used with small change in follow-up

Censoring time unknown in some individuals: Estimate time-to-censoring distribution. Then for those with competing event

- multiply impute censoring times
- reweight by probability to remain uncensored

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Left truncation Weights determined by time-to-entry distribution

Probabilistic setup data

- Event time T_i; may be right censored (administratively, loss to follow-up)
  Censoring time C_i
  $X_i = \min\{T_i, C_i\}$, $\Delta_i = I(T_i \leq C_i)$
- Left truncation (late entry).
  Entry time $V_i \sim \Phi$, $\Phi(t) = \mathrm{P}(V \leq t)$
- Sample: $\{V_i, X_i, E_i\Delta_i\}, i = 1, \ldots, N$
- Assume (V_i, C_i) independent of/non-informative for (T_i, E_i) (possibly conditional on covariables in regression model)

Estimation of $\overline{\Gamma}$ and Φ

Simplest: via standard PL-form

- $\widehat{\overline{\Gamma}}$: reverse role of event time T_i and censoring C_i:

$$\widehat{\overline{\Gamma}}(t) = \widehat{\mathrm{P}}(C > t) = \prod_{i: c_{(i)} \leq t} \left\{ 1 - \frac{m_i}{r(c_{(i)})} \right\}$$

  m_i: number of censorings at $c_{(i)}$

- $\widehat{\Phi}$: reverse role of $X_j = T_j \wedge C_j$ and truncation time V_j, V_j ("event") is right truncated by X_j:

$$\widehat{\Phi}(t) = \widehat{\mathrm{P}}(V \leq t) = \widehat{\mathrm{P}}(-V \geq -t)$$

Estimation of $\overline{\Gamma}$ and Φ

Simplest: via standard PL-form

- $\widehat{\overline{\Gamma}}$: reverse role of event time T_i and censoring C_i:

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- $\widehat{\Phi}$: reverse role of $X_j = T_j \wedge C_j$ and truncation time V_j, V_j ("event") is right truncated by X_j:

$$\widehat{\Phi}(t) = \widehat{\mathrm{P}}(V \leq t) = \widehat{\mathrm{P}}(-V \geq -t) = \prod_{i:-v_{(i)} < -t} \left\{ 1 - \frac{w_i}{r(v_{(i)})} \right\}$$

  w_i: number of entries at $v_{(i)}$

Estimation of $\overline{\Gamma}$ and Φ

Simplest: via standard PL-form

- $\widehat{\overline{\Gamma}}$: reverse role of event time T_i and censoring C_i:

$$\widehat{\overline{\Gamma}}(t) = \widehat{\mathrm{P}}(C > t) = \prod_{i:c_{(i)} \leq t} \left\{ 1 - \frac{m_i}{r(c_{(i)})} \right\}$$

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$$= \prod_{i:v_{(i)} > t} \left\{ 1 - \frac{w_i}{r(v_{(i)})} \right\}.$$

  w_i: number of entries at $v_{(i)}$

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Kaplan-Meier: product-limit formula

- Based on law of conditional probability

$$\mathrm{P}\{T > t\} = \prod_{i:t_{(i)} \le t} \mathrm{P}\{T > t_{(i)} | T > t_{(i)}-\}$$

$$= \prod_{i:t_{(i)} \le t} \left\{1 - h(t_{(i)})\right\}$$

- and estimate of hazard

$$\widehat{h}(t_{(i)}) = \frac{d(t_{(i)})}{r(t_{(i)})}$$

- Same for subdistribution

$$\mathrm{P}\{T_k > t\} = \prod_{i:t_{(i)} \le t} \mathrm{P}\{T_k > t_{(i)} | T_k > t_{(i)}-\}$$

$$= \prod_{i:t_{(i)} \le t} \left\{1 - h_k(t_{(i)})\right\}$$

General product-limit estimator of F_k

$$\widehat{F_k}^{\mathrm{PL}}(t) = \prod_{i:t_{(i)} \leq t} \left\{ 1 - \widehat{h_k}(t_{(i)}) \right\}$$

$$\widehat{h_k}(t_{(i)}) = \frac{d_k(t_{(i)})}{r^*(t_{(i)})}$$

General product-limit estimator of F_k

$$\widehat{F_k}^{\mathrm{PL}}(t) = \prod_{i:t_{(i)} \leq t} \left\{ 1 - \widehat{h_k}(t_{(i)}) \right\}$$

$$\widehat{h_k}(t_{(i)}) = \frac{d_k(t_{(i)})}{r^*(t_{(i)})}$$

Weights: contribution $\omega_l(t_{(i)})$ of individual l to $r^*(t_{(i)})$ is

- censored or event of type k before $t_{(i)}$: 0
- still at risk at $t_{(i)}$: 1

General product-limit estimator of F_k

$$\widehat{F_k}^{\mathrm{PL}}(t) = \prod_{i:t_{(i)} \leq t} \left\{ 1 - \widehat{h_k}(t_{(i)}) \right\}$$

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Weights: contribution $\omega_l(t_{(i)})$ of individual l to $r^*(t_{(i)})$ is

- censored or event of type k before $t_{(i)}$: 0
- still at risk at $t_{(i)}$: 1
- competing event at $t_{(j)}$ before $t_{(i)}$: weight

$$\frac{\widehat{\widehat{\Gamma}}(t_{(i)}-)}{\widehat{\widehat{\Gamma}}(t_{(j)}-)} \times \frac{\widehat{\Phi}(t_{(i)}-)}{\widehat{\Phi}(t_{(j)}-)}$$

General product-limit estimator of F_k

$$\widehat{\widehat{F_k}}^{\mathrm{PL}}(t) = \prod_{i:t_{(i)} \le t} \left\{ 1 - \widehat{h_k}(t_{(i)}) \right\}$$

$$\widehat{h_k}(t_{(i)}) = \frac{d_k(t_{(i)})}{r^*(t_{(i)})}$$

Weights: contribution $\omega_l(t_{(i)})$ of individual l to $r^*(t_{(i)})$ is

- censored or event of type k before $t_{(i)}$: 0
- still at risk at $t_{(i)}$: 1
- competing event at $t_{(j)}$ before $t_{(i)}$: weight

$$\frac{\widehat{\widehat{\Gamma}}(t_{(i)}-)}{\widehat{\widehat{\Gamma}}(t_{(j)}-)} \times \frac{\widehat{\Phi}(t_{(i)}-)}{\widehat{\Phi}(t_{(j)}-)}$$

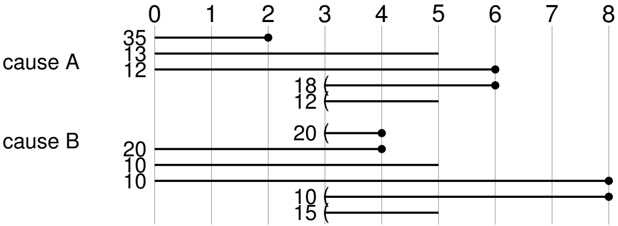
$$\approx \widehat{\mathrm{P}}\{C > t_{(i)} | C > t_{(j)}\} \times 1/\widehat{\mathrm{P}}\{V < t_{(j)} | V < t_{(i)}\}$$

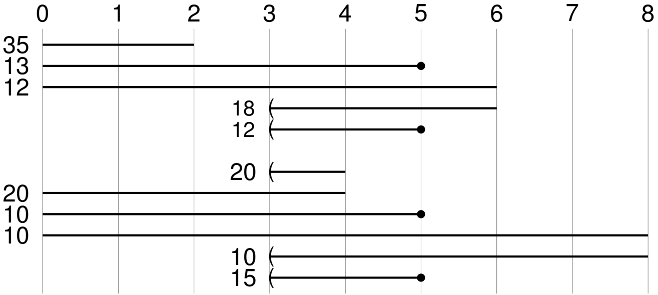
Equivalence

If weights in r^* based on the PL-form of $\widehat{\overline{\Gamma}}$ and $\widehat{\Phi}$, then we have

$$\widehat{F_k}^{\mathrm{AJ}} \equiv \widehat{F_k}^{\mathrm{PL}}$$

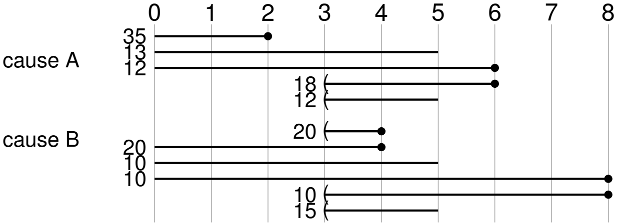
Calculation of $\widehat{\overline{\Gamma}}^{PL}$

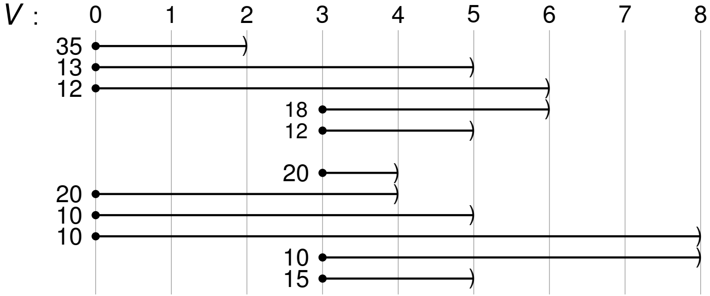




$\widehat{\overline{\Gamma}}^{PL}(t) = 1 - \frac{50}{100} = 0.5$ if $t \geq 5$; $\widehat{\overline{\Gamma}}^{PL}(t) = 1$ if $t < 5$

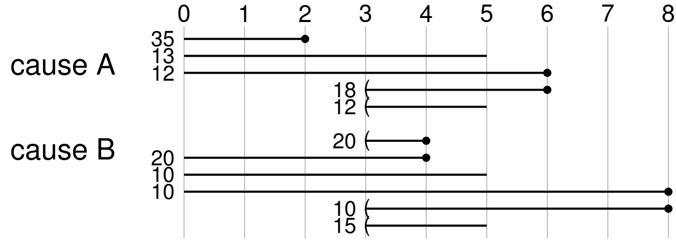
Calculation of $\widehat{\Phi}^{\mathrm{PL}}$





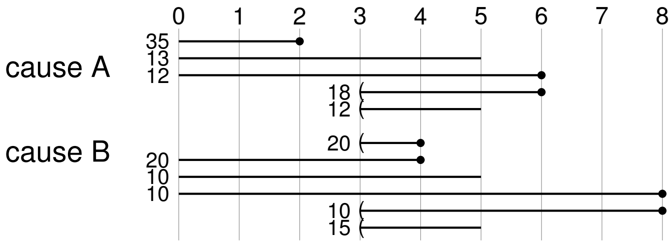
$\widehat{\Phi}(t) = 1 - \frac{75}{140} = \frac{65}{140}$ if $0 \le t < 3$; $\widehat{\Phi}(t) = 1$ if $t \ge 3$

Calculation of $\widehat{F_A}^{PL}(2)$ and $\widehat{F_A}^{PL}(6)$



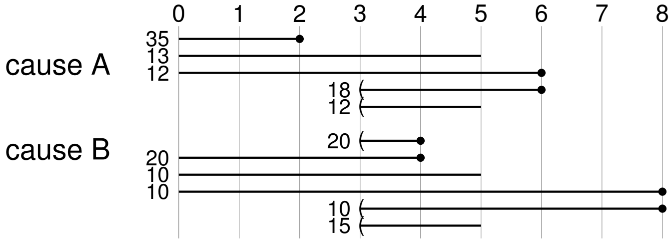
	2 years	4 years	6 years
$\widehat{\overline{\Gamma}}(t-)$	1	1	0.5
$\widehat{\Phi}(t-)$	$\frac{65}{140}$	1	1
$d(t)$	35 (A)	40 (B)	30 (A)

Calculation of $\widehat{F_A}^{PL}(2)$ and $\widehat{F_A}^{PL}(6)$



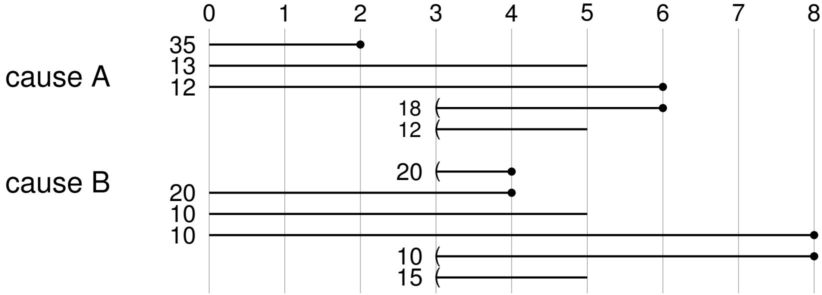
	2 years	4 years	6 years
$\widehat{\overline{\Gamma}}(t-)$	1	1	0.5
$\widehat{\Phi}(t-)$	$\frac{65}{140}$	1	1
$d(t)$	35 (A)	40 (B)	30 (A)
$r^*(t)$	100		$50 + 40 \times \frac{0.5}{1} \times \frac{1}{1} = 70$
$\widehat{h_A}(t)$	$\frac{35}{100}$	0	$\frac{30}{70}$

Calculation of $\widehat{F_A}^{PL}(2)$ and $\widehat{F_A}^{PL}(6)$



	2 years	4 years	6 years
$\widehat{\overline{\Gamma}}(t-)$	1	1	0.5
$\widehat{\Phi}(t-)$	$\frac{65}{140}$	1	1
$d(t)$	35 (A)	40 (B)	30 (A)
$r^*(t)$	100		$50 + 40 \times \frac{0.5}{1} \times \frac{1}{1} = 70$
$\widehat{h_A}(t)$	$\frac{35}{100}$	0	$\frac{30}{70}$
$\widehat{F_A}^{PL}(t)$	$(1 - \frac{35}{100}) = \frac{65}{100}$	$\frac{65}{100}$	$(1 - \frac{35}{100}) \times (1 - \frac{30}{70}) = \frac{13}{35}$

Detailed calculation of $\widehat{F_A}^{\mathrm{AJ}}(2)$ and $\widehat{F_A}^{\mathrm{AJ}}(6)$



	2 years	4 years	6 years
$\widehat{\lambda_{\mathrm{A}}}(t)$	$\frac{35}{100}$	0	$\frac{30}{50}$
$\widehat{h}(t)$	$\frac{35}{100}$	$\frac{40}{140}$	$\frac{30}{50}$
$\widehat{\mathrm{P}}(T \geq t)$	1	$(1 - \frac{35}{100}) = \frac{65}{100}$	$\frac{65}{100} \times (1 - \frac{40}{140}) = \frac{65}{140}$
$\widehat{F_A}^{\mathrm{AJ}}(t)$	$1 \times \frac{35}{100} = \frac{35}{100}$	$\frac{35}{100} + \frac{65}{100} \times 0 = \frac{35}{100}$	$\frac{35}{100} + \frac{65}{140} \times \frac{30}{50} = \frac{22}{35}$

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Standard error/confidence intervals

Based on formulas from Kaplan-Meier

Standard error/confidence intervals

Based on formulas from Kaplan-Meier

- Based on estimator of subdistribution hazard instead of cause-specific hazard.

$$\widehat{\text{var}}\left\{\widehat{H_k}(t)\right\} = \widehat{\text{var}}\left\{\sum_{t_{(j)}\leq t} \frac{d_k(t_{(j)})}{r^*(t_{(j)})}\right\}$$

Two forms: Aalen and Greenwood.

Standard error/confidence intervals

Based on formulas from Kaplan-Meier

- Based on estimator of subdistribution hazard instead of cause-specific hazard.

$$\widehat{\text{var}}\left\{\widehat{H_k}(t)\right\} = \widehat{\text{var}}\left\{\sum_{t_{(j)}\leq t} \frac{d_k(t_{(j)})}{r^*(t_{(j)})}\right\}$$

  Two forms: Aalen and Greenwood.

- Confidence intervals: e.g. on log-scale (via $\log\{\widehat{\overline{F_k}}(t) \approx -\widehat{H_k}(t)\}$):

$$\exp\left[\log\left\{\widehat{\overline{F_k}}(t)\right\} \pm k_\alpha\, \widehat{\text{s.e.}}\{\widehat{H_k}(t)\}\right]$$

Standard error/confidence intervals

Based on formulas from Kaplan-Meier

- Based on estimator of subdistribution hazard instead of cause-specific hazard.

$$\widehat{\mathrm{var}}\left\{\widehat{H_k}(t)\right\} = \widehat{\mathrm{var}}\left\{\sum_{t_{(j)}\leq t} \frac{d_k(t_{(j)})}{r^*(t_{(j)})}\right\}$$

  Two forms: Aalen and Greenwood.

- Confidence intervals: e.g. on log-scale (via $\log\{\widehat{F_k}(t) \approx -\widehat{H_k}(t)\}$):

$$\exp\left[\log\left\{\widehat{F_k}(t)\right\} \pm k_\alpha\, \widehat{\mathrm{s.e.}}\{\widehat{H_k}(t)\}\right]$$

- PL-form seems to perform as well as approach based on AJ-form in most settings

Estimators in competing risks setting

hazard	estimate	cumulative
marginal	$d_k(t)/r(t)\{=\widehat{\lambda_k}(t)\}$	$\widehat{F^m}^{\text{PL}} = \prod_{t_{(j)} \leq t} \left\{1 - \widehat{\lambda_k}(t_{(j)})\right\}$
cause-specific	$\widehat{\lambda_k}(t) = d_k(t)/r(t)$	$\widehat{F_k}^{\text{AJ}}(t) = \sum_{t_{(j)} \leq t} \widehat{\overline{F}}^{\text{PL}}(t_{(j)}-)\widehat{\lambda_k}(t_{(j)})$
subdistribution	$\widehat{h_k}(t) = d_k(t)/r^*(t)$	$\widehat{\overline{F_k}}^{\text{PL}}(t) = \prod_{t_{(j)} \leq t} \left\{1 - \widehat{h_k}(t_{(j)})\right\}$
overall	$\widehat{h}(t) = d(t)/r(t)$	$\widehat{\overline{F}}^{\text{PL}}(t) = \prod_{t_{(j)} \leq t} \left\{1 - \widehat{h}(t_{(j)})\right\}$

Competing risks

- Competing risk is treated as a separate event
  - Individuals censored by competing event don't have to be represented by the ones that remain at risk.
    Other censoring (administrative/loss to follow-up) must be independent
- Cause-specific hazard
  - Basis for Aalen-Johansen estimator of cause-specific cumulative incidence/crude risk
  - If censoring due to competing event is independent, then marginal and cause-specific hazard are equal. Cumulative quantities different: Kaplan-Meier versus Aalen-Johansen
- Subdistribution hazard: one-to-one relation with crude risk.
  Product-limit estimator equal to Aalen-Johansen estimator with our choice of weights

Some relations and properties

- Estimate of overall hazard is sum of cause-specific hazard estimates

$$\frac{\sum_{e=1}^{K} d_e(t_{(i)})}{r(t_{(i)})} = \frac{d(t_{(i)})}{r(t_{(i)})}$$

Some relations and properties

- Estimate of overall hazard is sum of cause-specific hazard estimates

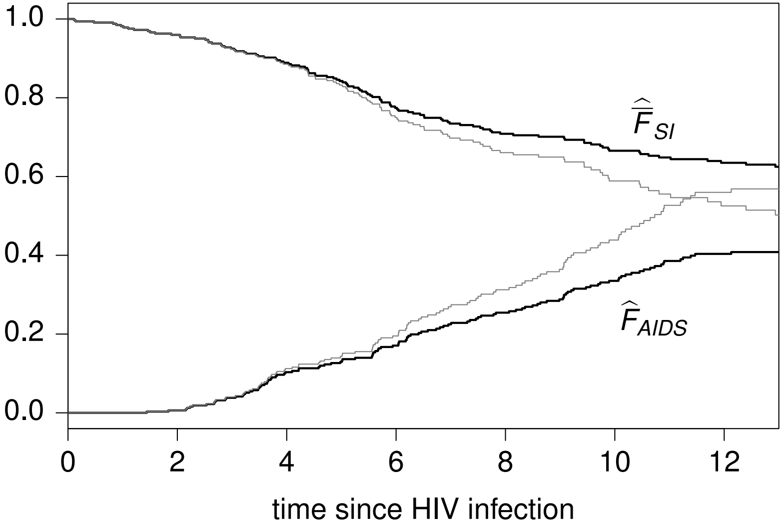
$$\frac{\sum_{e=1}^{K} d_e(t_{(i)})}{r(t_{(i)})} = \frac{d(t_{(i)})}{r(t_{(i)})}$$

- Sum of Aalen-Johansen (or weighted PL) estimates is Kaplan-Meier of overall event risk

Some relations and properties

- Estimate of overall hazard is sum of cause-specific hazard estimates
$$\frac{\sum_{e=1}^{K} d_e(t_{(i)})}{r(t_{(i)})} = \frac{d(t_{(i)})}{r(t_{(i)})}$$
- Sum of Aalen-Johansen (or weighted PL) estimates is Kaplan-Meier of overall event risk
- Marginal Kaplan-Meier always steeper than weighted PL (or Aalen-Johansen) estimate, because
$$r(t_{(i)}) \leq r^*(t_{(i)})$$

Alternate display format



Some relations and properties

- Estimate of overall hazard is sum of cause-specific hazard estimates

$$\frac{\sum_{e=1}^{K} d_e(t_{(i)})}{r(t_{(i)})} = \frac{d(t_{(i)})}{r(t_{(i)})}$$

- Sum of Aalen-Johansen (or weighted PL) estimates is Kaplan-Meier of overall event risk
- Marginal Kaplan-Meier always steeper than weighted PL (or Aalen-Johansen) estimate, because

$$r(t_{(i)}) \leq r^*(t_{(i)})$$

- Relation between both hazards

$$\widehat{\lambda_k}(t) \times \widehat{\overline{F}}^{\mathrm{PL}}(t-) = \widehat{h_k}(t) \times \widehat{\overline{F_k}}^{\mathrm{PL}}(t-)$$

Outline

Subdistribution
- Subdistribution hazard
- Product-limit estimator
- Confidence intervals based on PL-estimator

Log-rank tests

Proportional subdistribution hazards
- Standard errors

Testing for differences between curves

- Log-rank test is a nonparametric test for equality of hazards over groups. One of family of tests. E.g. with two groups in classical setting:

$$Z = \sum_{j=1}^{n} W(t_{(j)}) \frac{r^{(1)}(t_{(j)}) r^{(2)}(t_{(j)})}{r^{(1)}(t_{(j)}) + r^{(2)}(t_{(j)})} \left[\frac{d^{(1)}(t_{(j)})}{r^{(1)}(t_{(j)})} - \frac{d^{(2)}(t_{(j)})}{r^{(2)}(t_{(j)})} \right]$$

Log-rank: $W \equiv 1$

Three types of hazards

1. marginal, only estimable under independence, conditionally on the covariables in the model
2. cause-specific; estimable (standard censoring) No 1-1 relation with cause specific cumulative incidence
3. subdistribution; estimable (weighted risk set) 1-1 relation with cause specific cumulative incidence

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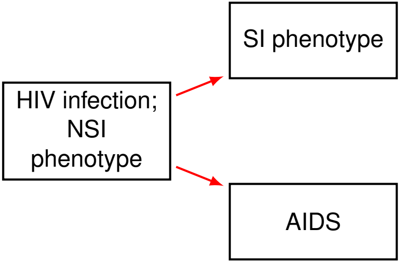
- Standard log-rank test (one event type): nonparametric comparison of marginal hazards $\rightarrow$ compares marginal time-to-event distributions (1-1 relation)
- Standard log-rank test (competing risks): nonparametric comparison of cause-specific hazards
- Gray's log-rank test: replace r by r^*. Nonparametric test to compare subdistribution hazards $\rightarrow$ compares cause-specific cumulative incidences (1-1 relation)

C: AIDS and SI switch after HIV infection



- NSI virus needs CCR5 co-receptor for cell entry
- Gene coding for CCR5 may have deletion CCR5-Δ32 on one chromosome
- Individuals with CCR5-Δ32 have slower AIDS progression

- SI: “syncytium inducing phenotype”. At HIV infection only NSI
- Competing risks: time to either SI switch or AIDS as first event

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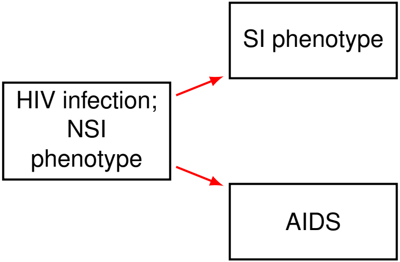

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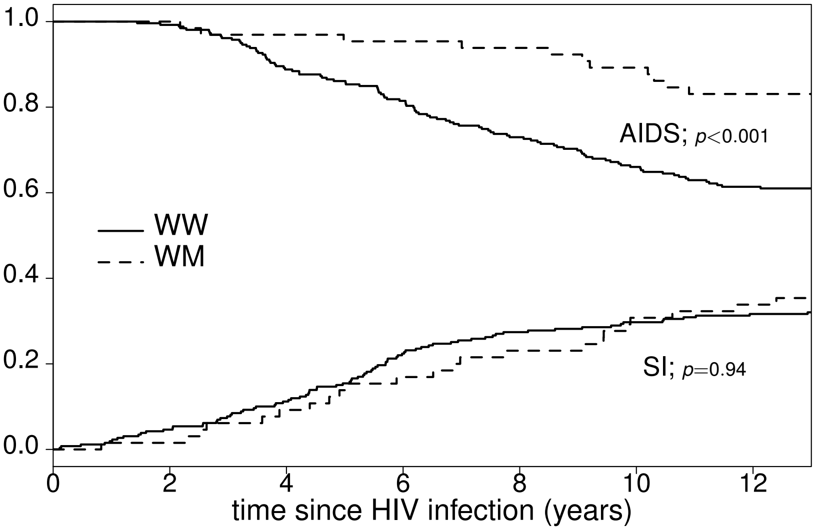
- Question: SI switch as first event more likely for CCR5-Δ32 individuals?

Estimates by value of CCR5, with Gray's log-rank test



Exercise III.14.1

When there are no competing risks, a log-rank test is used to compare Kaplan-Meier cumulative incidence curves. In competing risks settings, this test is inappropriate.

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Fine and Gray model

- Regression on subdistribution hazard

$$h_k(t \mid \mathbf{Z}_i) = h_{k,0}(t)\ \exp(\boldsymbol{\beta}_k^\top \mathbf{Z}_i)$$

- Original idea by Fine and Gray (1999)
  Sometimes called “competing risks proportional hazards”
- Estimation based on weighted partial likelihood:

$$L(\boldsymbol{\beta}_k) = \prod_{\substack{i=1 \\ e_i \delta_i = k}}^{N} \left\{ \frac{\exp(\boldsymbol{\beta}_k^\top \mathbf{Z}_i)}{\sum_{j=1}^{N} \omega_j(t_i) \exp(\boldsymbol{\beta}_k^\top \mathbf{Z}_j)} \right\}. \tag{1}$$

Test for effect of CCR5-Δ32 deletion

- Cause-specific hazards (WM versus WW)

	HR	p-value (Cox)	log-rank	df
AIDS	0.291	<0.001	<0.001	1
SI	0.776	0.29	0.28	1

Fitted cumulative incidence curves to SI cross

Test for effect of CCR5-Δ32 deletion

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	HR	p-value (Cox)	log-rank	df
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  Fitted cumulative incidence curves to SI cross

- Subdistribution:

	HR	p-value (F&G)	log-rank	df
AIDS	0.366	<0.001	<0.001	1
SI	1.024	0.92	0.94	1

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Standard errors

- Fine & Gray use sandwich type estimator
  Also default in `coxph` in recent versions of `survival`
- In simulation study, standard estimator performed better:
  - Two covariables, Z_1 and $Z_2 \sim N(0,1)$, effect $\beta_1 = \beta_2 = 0.5$
  - T° unit exponential with mass 0.7 at ∞ if $Z_1 = Z_2 = 0$
  - Censoring uniform on $[0.5, 1]$

Results for β_1 (5000 runs):

	N	bias	p-value
`coxph` with `robust =FALSE`	100	0.018	<0.001
	200	0.005	0.043
	500	0.001	0.38
`crr` or `coxph`	100	0.018	<0.001
	200	0.005	0.043
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	N	bias	p-value	$\widehat{\text{var}(\beta_1)}$	$\overline{\text{var}(\beta_1)}$
`coxph` with	100	0.018	<0.001	0.069	0.065
`robust`	200	0.005	0.043	0.032	0.030
`=FALSE`	500	0.001	0.38	0.012	0.011
`crr` or	100	0.018	<0.001	0.069	0.062
`coxph`	200	0.005	0.043	0.032	0.029
	500	0.001	0.38	0.012	0.011

Standard errors

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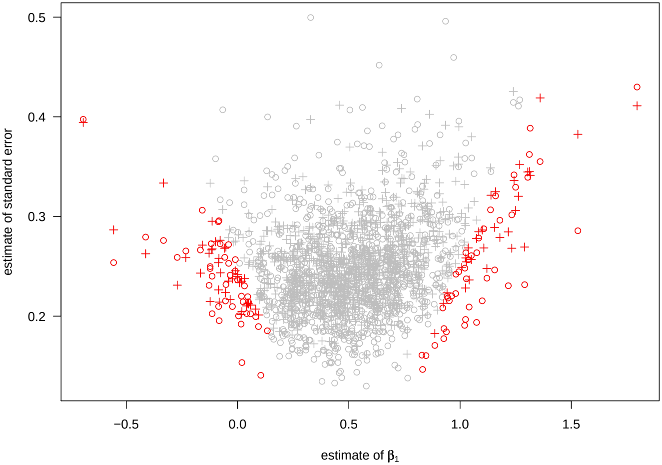
Results for β_1 (5000 runs):

	N	bias	p-value	$\widehat{\mathrm{var}(\beta_1)}$	$\overline{\mathrm{var}(\beta_1)}$	$P(0.5 \in \mathrm{CI})$	p-value
`coxph` with	100	0.018	<0.001	0.069	0.065	0.946	0.21
`robust`	200	0.005	0.043	0.032	0.030	0.944	0.074
`=FALSE`	500	0.001	0.38	0.012	0.011	0.950	1.00
`crr` or	100	0.018	<0.001	0.069	0.062	0.928	<0.001
`coxph`	200	0.005	0.043	0.032	0.029	0.934	<0.001
	500	0.001	0.38	0.012	0.011	0.945	0.14

Standard errors



Why Does Sandwich Perform Worse?

- Sandwich asymptotically unbiased
- Censoring characteristics in sample is what matters, not the theoretical distribution of the time-to-censoring distribution
- Only individuals without event are reweighted in ω_i-weights.

Exercise 16.1

In patients that receive allogeneic hematopoietic stem cell transplantation, relapse and treatment related mortality (TRM) are two competing events. Comment on the following statements:

I. The effect of a covariable on an event from either a cause-specific (e.g. relapse) model or overall (e.g. relapse and TRM combined) model may be very different from the effect of the covariable on the event (e.g. relapse) in the presence of competing risks.

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II. Using a cause-specific Cox regression model is incorrect because it ignores competing risks and treats them as censored.